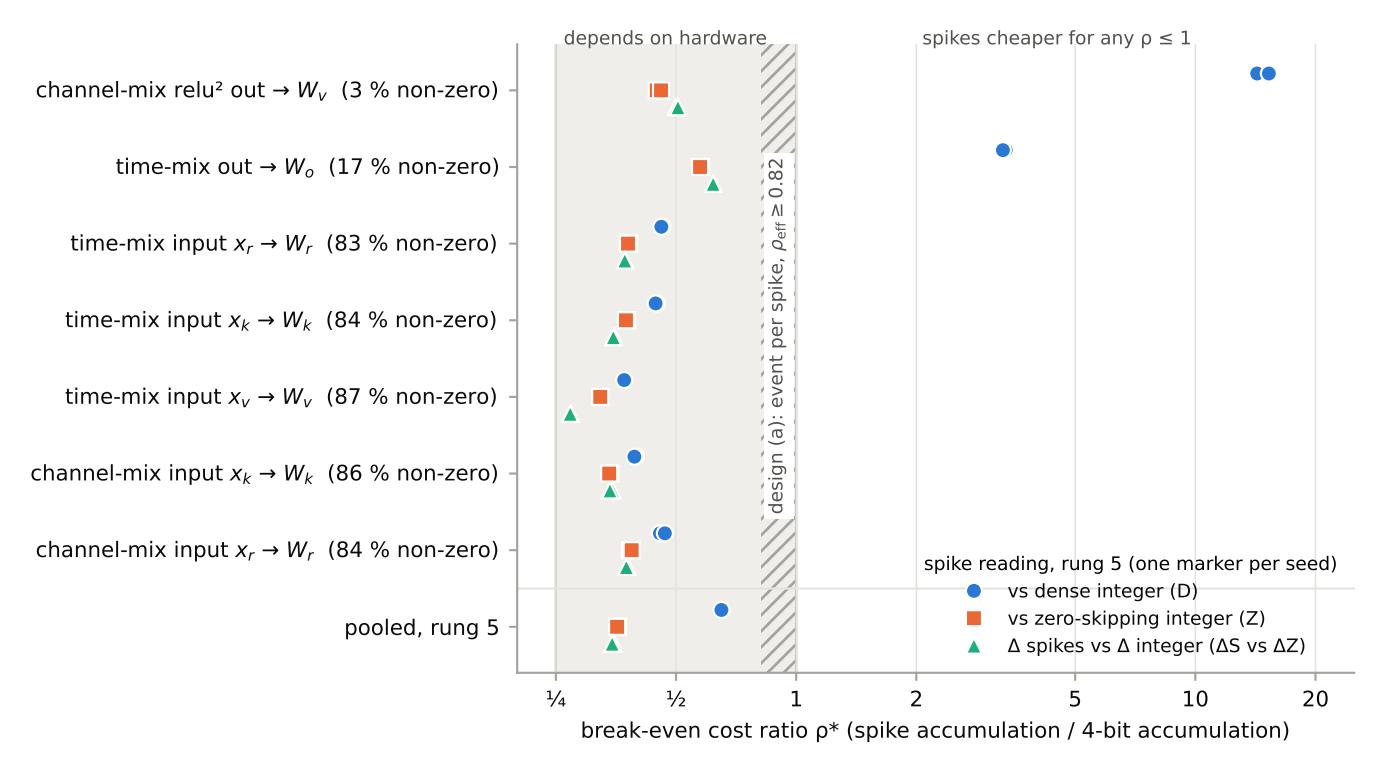
depends on hardware
spikes cheaper for any ρ ≤ 1
design (a): event per spike, $\rho_{\text{eff}} \geq 0.82$
channel-mix relu² out → W_v (3 % non-zero)
time-mix out → W_o (17 % non-zero)
time-mix input x_r → W_r (83 % non-zero)
time-mix input x_k → W_k (84 % non-zero)
time-mix input x_v → W_v (87 % non-zero)
channel-mix input x_k → W_k (86 % non-zero)
channel-mix input x_r → W_r (84 % non-zero)
pooled, rung 5
spike reading, rung 5 (one marker per seed)
vs dense integer (D)
vs zero-skipping integer (Z)
Δ spikes vs Δ integer (ΔS vs ΔZ)
¼
½
1
2
5
10
20
break-even cost ratio ρ* (spike accumulation / 4-bit accumulation)